

Question

The laboratory determines the content of titanium and iron in ilmenite containing only inert impurities using the following method:

(1) Accurately weigh a 0.4278 g sample of ilmenite into a 30 mL porcelain crucible, add 10 g potassium pyrosulfate, and fuse at 700-750°C for decomposition. After cooling, leach the fused mass into a conical flask using 80 mL of 1:1 hydrochloric acid and 20 mL of 1:1 sulfuric acid in portions. Rinse the crucible with a small amount of hot water and transfer the washings to the same conical flask.

(2) Add 3 g aluminum foil to the conical flask containing the leachate, and heat on an electric hotplate while slowly bubbling carbon dioxide until vigorous bubbling occurs. Cool to room temperature, add 10 mL of 400 g/L ammonium thiocyanate solution, and titrate with standard ferric ammonium sulfate solution until a blood-red color indicates the endpoint. The consumption of ferric ammonium sulfate solution is 25.49 mL.

(3) Weigh 0.2174 g of high-purity titanium dioxide, dissolve it in acid, and treat it under the same two-step conditions described above. Titrate with standard ferric ammonium sulfate solution, consuming 26.79 mL.

The correct options are (error less than 1% is considered correct):

A. The titanium content in the mineral is 15.46%.

B. The titanium content in the mineral is 24.83%.

C. The titanium content in the mineral is 29.44%.

D. The titanium content in the mineral is 35.75%.

E. The iron content in the mineral is 20.99%.

F. The iron content in the mineral is 26.30%.

G. The iron content in the mineral is 35.88%.

H. The iron content in the mineral is 40.22%.

I. None of the above options is correct

Answer

Correct Answer: **I**

Detailed Explanation

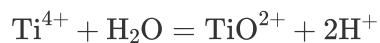
Ilmenite generally refers to FeTiO_3 ; when fused with potassium pyrosulfate, it forms the highest-valence sulfates, namely $\text{Fe}_2(\text{SO}_4)_3$, $\text{Ti}(\text{SO}_4)_2$.

CHECKPOINT

1 PTS

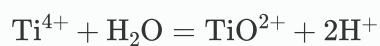
FeTiO_3 forms $\text{Fe}_2(\text{SO}_4)_3$, $\text{Ti}(\text{SO}_4)_2$ when fused with potassium pyrosulfate

Ti^{4+} is unstable in solution and hydrolyzes to TiO^{2+} :

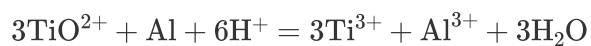


CHECKPOINT

1 PTS

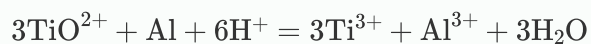


TiO^{2+} is strongly oxidizing and can oxidize elemental aluminum:

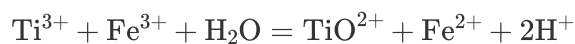


CHECKPOINT

1 PTS

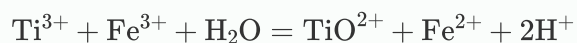


The resulting Ti^{3+} is then converted back to TiO^{2+} by Fe^{3+} in the system:



CHECKPOINT

1 PTS



Thus, it is equivalent to Fe^{3+} oxidizing aluminum first, followed by TiO^{2+} oxidizing aluminum.

CHECKPOINT

1 PTS

It is equivalent to Fe^{3+} oxidizing aluminum first, followed by TiO^{2+} oxidizing aluminum

Excess aluminum is oxidized by H^+ .

CHECKPOINT

1 PTS

Excess aluminum is oxidized by H^+ .

Ultimately, the system contains only Fe^{2+} and Ti^{3+} , and ammonium ferric sulfate standard solution is added to titrate Ti^{3+} to the endpoint.

The standard solution of ammonium ferric sulfate is calibrated using pure titanium dioxide, with the reaction equation being the same as above and the stoichiometric ratio being 1:1, thus:

$$c(\text{Fe}^{3+}) = 0.2174/79.87/0.002679 = 0.1016 \text{ mol/L}$$

CHECKPOINT

1 PTS

The concentration of the ammonium ferric sulfate standard solution $c(\text{Fe}^{3+}) = 0.1016 \text{ mol/L}$

The titanium content in the sample can be calculated as:

$$\omega(\text{Ti}) = 0.1016 \times 0.02549 \times 47.87/0.4278 \times 100\% = 28.99\%$$

CHECKPOINT

1 PTS

$$\omega(\text{Ti}) = 28.99\%$$

Based on the chemical formula of ilmenite and its inert impurities, the iron content is:

$$\omega(\text{Fe}) = 0.1016 \times 0.02549 \times 55.85/0.4278 \times 100\% = 33.81\%$$

CHECKPOINT

1 PTS

$$\omega(\text{Fe}) = 33.81\%$$

Consequently, options A-H are all incorrect, and option I is correct.